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Timing of rotational motion of Southwest Japan inferred from paleomagnetism

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Paleomagnetic field directions for the period younger than 35 Ma are obtained from igneous rocks distributed in the San'in district, Inner Zone of Southwest Japan. The remanent magnetization of samples between 30 and 35 Ma old are fairly well grouped with a mean direction of $D = 65.9°$, $I = 48.6°$, and $\alpha_{95} = 6.5°$. This result establishes that Southwest Japan rotated clockwise $56 \pm 12°$ during the past 30 m.y. A declination value of about 60° is observed in the rocks of 28 Ma ($D = 52.2°$, $I = 33.5°$) and 21 Ma ($D = 69.9°$, $I = 49.5°$). Comparing this with results from dacitic rocks with an age of 15 Ma in other areas of Southwest Japan suggests that rotational motion did not occur possibly until 15 Ma. These results require that the rotation of Southwest Japan occurred just after the Shikoku Basin had been created.

1. Introduction

The rotation of island arcs in the Tertiary period have been inferred from paleomagnetic studies in the western Pacific. Southwest Japan, the northern arm of Sulawesi, Panay in the central Philippines and the Southern Mariana Islands are suggested to have been subjected to clockwise rotation [1–5], while Luzon island preserves a record of counterclockwise rotation [6]. These rotations are apparently attributed to either the tectonics associated with back-arc spreading (e.g. Southwest Japan) or collision between land blocks (Panay, Luzon, northern arm of Sulawesi). The rotational motion is also a reflection of the mechanical behaviour of the lithosphere produced by forces imparted from the asthenosphere. An investigation of the rotational motion leads ultimately to an understanding of the mechanical interaction between the lithosphere and asthenosphere as well as the kinematics of island arcs.

A detailed description of the rotation is required to study the mechanism of rotational motion of island arcs from a theoretical viewpoint. Southwest Japan provides an opportunity to improve the resolution of rotational motion. Since a short sketch of the rotational motion in the Tertiary was given by Otofuji and Matsuda [1], extensive paleomagnetic work combined with age determinations documents the full process of rotation.

The work presented here is an attempt to determine the timing of rotation of Southwest Japan based on more detailed paleomagnetic data. Our paleomagnetic approach is to measure the natural remanent magnetization (NRM) of rocks between 30 Ma and 35 Ma old. These data provide the declination value which characterizes the amount of rotation of Southwest Japan, because the main part of this rotation occurred later than this period [1]. The time interval of 5 Ma in this period is small enough to introduce no effect of apparent polar wander, because swift apparent polar wander ceased by 40 Ma [7,8]. To estimate the timing of

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the start of rotation, the declination value of 30 to 35 Ma is compared with those of rocks younger than 30 Ma.

A reliable rotation angle about the vertical axis is obtained from the declination of rocks after tilting correction. The greatest source of error in inferring the declination value is the uncertainty in making the correction for tectonic deformation. The paleomagnetic work in this paper is focused only on rocks which satisfy either of the following conditions; (1) geological evidence indicates that the rocks have been subjected to little tectonic tilting, or (2) tilting of the rocks can be corrected. These rocks yield fairly reliable data.

2. Geology and sampling

Extensive late Paleogene eruptives occurred in the San'in district, Inner Zone of Southwest Japan.

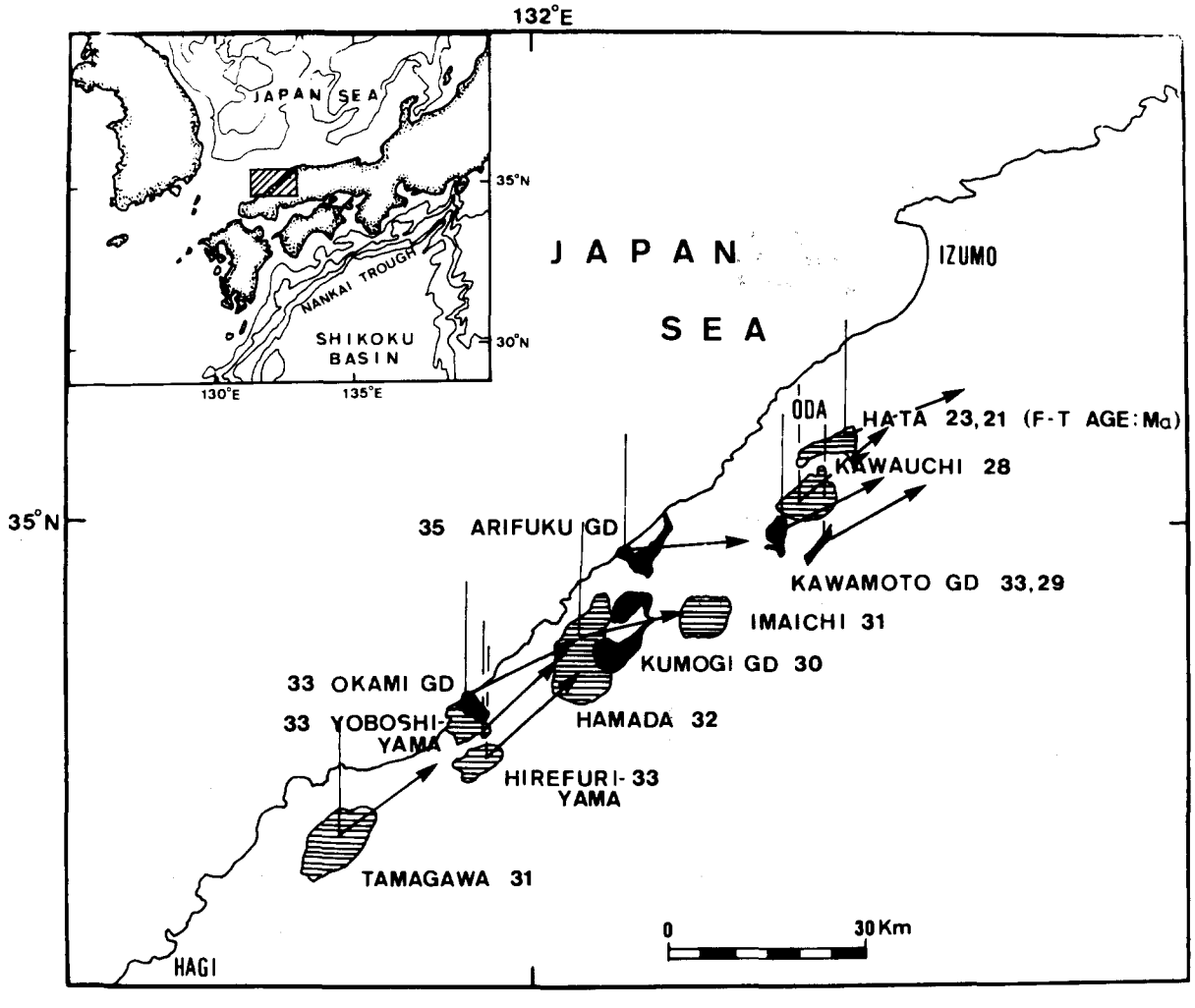


Fig. 1. Locations of rock units where samples were collected. The shaded rock unit marks the volcanic production associated with cauldron formation and the solid rock unit the plutonic stock. Sampling localities are listed in Table 1. Arrows indicate the declinations of mean paleomagnetic directions obtained for the respective rock units. The age obtained by fission-track dating [12] is assigned to each rock unit.

The volcanism was associated with cauldron subsidence; the volcanic products at each cauldron formed a rock unit which consisted of rhyodacitic to andesitic lavas and rhyodacitic to dacitic pyroclastics. Geological work has shown cauldron locations and stratigraphy [9–11]. Plutonic rocks were intruded contemporaneously in this district. These volcanic and plutonic rocks have been dated by the fission-track dating tecnique [12]; their ages range from 28 to 35 Ma. Continued dacitic to andesitic volcanism produced a younger volcanic unit (Hata Subgroup) whose age was estimated to be 21–23 Ma by fission-track dating [12]. These volcanic units and the plutonic rocks in this district were chosen for our work (Fig. 1).

Most samples were collected from the welded tuff in pyroclastics of each rock unit, because welded tuff is very suitable for paleomagnetism; a reliable paleomagnetic direction could be obtained [13,14] and the eutaxitic structure (lineation of stretched pumice and aligned phenocryst) is a good indicator of the horizontal plane at the time of eruption [15]. Sampling was carried out at outcrops in which tilting was clearly observed either through the eutaxitic structure of welded tuff, or by sediment layers within or above andesite lavas. Outcrops with different tilting were preferred for sampling sites in each rock unit. This allowed us to evaluate the effect of the tilting correction. Samples were also collected from plutonic rocks which were estimated to have undergone little tilt. The sampling locations in each stock were widely spaced to cancel out the effect of tilting.

More than 400 samples were collected from seven volcanic units at 26 sites and from four plutonic rocks at 10 sites (Fig. 1). Sampling localities are listed in Table 1. The samples were taken by hand; a magnetic compass was used for orientation. Each site typically comprised twelve independently oriented samples distributed over distances ranging up to 20 m.

3. Paleomagnetism

Individual specimens, 25 mm in diameter and 25 mm long, were prepared from samples in the laboratory. A Schonstedt SSM-1A spinner magnetometer was used to measure the remanent magnetization of the specimens. The stability of remanence was investigated through progressive demagnetization treatment by both thermal and alternating field (AF) methods. Three pilot specimens from each site were thermally demagnetized in more than six steps. Specimens were heated in air in a non-inductively wound electric furnace enclosed in three cylindrical mu-metal shields in which the residual magnetic field was less than 10 nT. Demagnetization in alternating fields up to 50 mT was carried out on one pilot specimen from each site with an magnetically shielded two-axis tumbler. Stability of remanence to thermal demagnetization for each site is given in Table 1. A magnetically stable site is defined as follows: the change in NRM directions of pilot specimens was less than 10° during thermal demagnetizations. Typical results of progressive thermal demagnetization are shown in Fig. 2.

3.1. Results of demagnetization

One specimen from all independently oriented samples for each site were treated with thermal demagnetization to identify the primary component, because we found that the thermal demagnetization technique was more successful in erasing the secondary component than was AF demagnetization. The optimum demagnetization temperature to produce minimum dispersion was selected from three pilot specimens for magnetically stable sites. All samples for the other sites were progressively demagnetized to define the thermally stable end point of the magnetization. The range of demagnetization level required to determine the characteristic direction is listed in Table 1.

Granodiorite and andesite. Internally consistent data were obtained after thermal demagnetization from these rocks. The most striking results were observed in granodiorite sites IM 314 and IM 323 (Fig. 3). Scattered NRM directions with precision parameter k [16] of less than 5 made tight clusters after demagnetization above 500°C; the precision parameter increased up to 60 at site IM 323. The thermal demagnetization curves of granodiorite and andesite rocks revealed a Curie point of 580°C

TABLE 1

Summary of paleomagnetic results

Rock unit	Age (Ma)	Site	Locality	Demagnetization		N	D(°)	I(°)	α_{95}(°)	k
Hata										
Uppermost andesite lava		IM 303	132°30.9′, 35°06.3′	500–560 °C	B	10	−109.1	−57.6	7.3	44.6
Upper rhyodacite welded tuff	21 ± 1	IM 301	132°30.6′, 35°05.6′	400 °C	A	10	86.3	25.3	2.0	605.4
		IM 302	132°31.3′, 35°05.7′	400 °C	A	12	61.9	56.1	1.9	532.5
		IM 417	132°31.0′, 35°05.3′	400 °C	A	12	73.9	44.3	3.4	161.8
		IM 418	132°30.6′, 35°05.2′	400 °C	A	12	46.8	36.7	3.0	206.2
Lower andesite lava	23	IM 202 *	132°32.6′, 35°05.7′	550 °C	B	9	−104.2	−64.1	6.1	72.3
Kawauchi (Sasahara Fm.)										
Dacite welded tuff	28	IM 107 *	132°28.0′, 34°59.5′	400 °C	A	10	52.4	15.9	3.6	185.6
		IM 212 *	132°25.7′, 35°00.8′	500 °C	A	11	83.5	21.6	5.8	63.2
		IM 213 *	132°28.4′, 35°00.2′	500 °C	A	12	42.0	30.9	1.8	552.1
		IM 214 *	132°28.5′, 34°59.5′	500 °C	A	11	35.6	35.1	3.6	164.1
		IM 421	132°27.4′, 34°58.7′	400 °C	A	12	48.6	35.0	2.2	376.7
		IM 422	132′28.4′, 35°00.3′	400 °C	A	12	43.6	47.3	3.6	147.9
Imaichi										
Rhyodacite welded tuff	31 ± 2	IM 321	132°18.0′, 34°52.1′	400 °C	B	11	130.6	38.5	2.5	322.3
		IM 322	132°15.8′, 34°52.6′	400 °C	A	8	−104.6	33.3	6.2	79.8
Hamada										
Rhyodacite welded tuff	32 ± 1	IM 316	132°07.5′, 34°52.0′	600–640 °C	B	10	−108.9	−31.7	4.5	117.2
		IM 317	132°07.1′, 34°52.7′	500–540 °C	A	13	98.6	43.2	2.1	378.0
		IM 412	132°06.9′, 34°52.6′	150 °C	B	12	−134.9	−38.3	5.1	72.7
		IM 413	132°07.3′, 34°52.1′	400 °C	A	13	83.8	33.2	4.3	93.9
		IM 414	132°07.2′, 34°51.7′	400 °C	A	13	89.7	25.9	3.9	113.9
Hirefuriyama										
Rhyodacite welded tuff	33 ± 2	IM 405	131°56.7′, 34°41.2′	400 °C	A	11	−117.3	−37.1	3.4	185.7
		IM 406	131°56.6′, 34°41.6′	400 °C	A	11	−136.9	−32.1	3.4	180.7
Andesite lava		IM 407	131°52.7′, 34°40.7′	400 °C	A	12	−149.4	−68.1	4.1	110.5
		IM 408	131°52.9′, 34°40.9′	500 °C	B	12	−132.9	−43.9	3.3	169.1
Yoboshiyama										
Rhyodacite welded tuff	33 ± 2	IM 313	131°55.0′, 34°44.0′	400 °C	A	12	46.9	30.4	3.8	129.5
Andesite porphyrite		IM 312	131°55.5′, 34°43.3′	400–500 °C	B	11	44.1	33.3	10.0	21.7
Tamagawa										
Dacite welded tuff	31 ± 1	IM 404	131°41.0′, 34°36.8′	500 °C	B	5	−123.8	−60.1	6.6	134.0
Okami										
Granodiorite	33 ± 2	IM 314	131°54.4′, 34°46.3′	520–560 °C	B	12	79.2	57.9	8.5	27.1
		IM 410	131°54.1′, 34°46.2′	500 °C	B	9	71.2	63.5	7.3	51.1
		IM 411	131°54.0′, 34°45.3′	500 °C	B	8	45.3	53.0	16.1	12.8
Arifuku										
Granodiorite	35 ± 2	IM 323	132°10.8′, 34°57.0′	520–540 °C	B	10	−127.5	−55.4	6.3	60.0
		IM 324	132°09.1′, 34°58.1′	520–560 °C	B	10	−118.6	−37.7	5.6	75.4
		IM 415	132°11.7′, 34°59.5′	300–500 °C	B	11	−74.4	−44.1	9.9	22.4
		IM 416	132°14.0′, 35°00.2′	400–500 °C	B	8	−69.0	−29.0	9.3	36.4

All data except for those of granodiorite are after tilting correction. Data denoted by asterisk (*) are from Otofuji and Matsuda [1]. The ages are fission-track data from Matsuda [12]. Demagnetization: the magnetic cleaning temperature and the behaviour of NRM direction during thermal demagnetization. A = (stable); change in NRM direction was less than 10° during demagnetization; B = (unstable); change was more than 10°. N is the number of samples in site mean computation, D is the site mean declination, I is the site mean inclination, α_{95} is the radius of the 95% circle of confidence around the site mean direction and k is Fisher precision parameters.

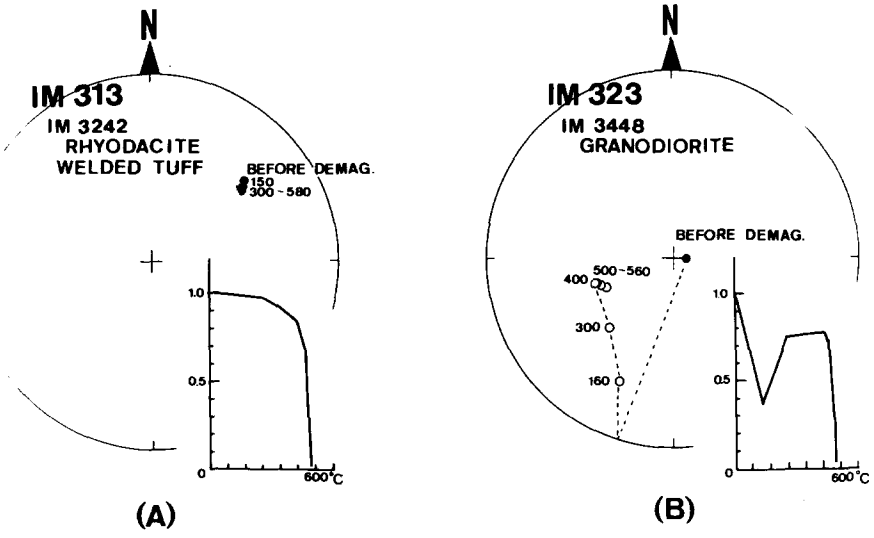


Fig. 2. Thermal demagnetization of a magnetically stable sample from welded tuff in Yoboshiyama rock unit (IM 313) (A) and a unstable one from Arifuku granodiorite (IM 323) (B). Most welded tuff samples are magnetically stable. Specimens from the granodiorite and andesite changed in direction during thermal demagnetization, and their directions reached the stable end point at above 300–500 °C. The increase in intensity was sometimes observed during demagnetization at 150–300 °C, suggesting the removal of overprinting component. Projections are equal area, solid symbols on the lower hemisphere, open symbols on the upper hemisphere.

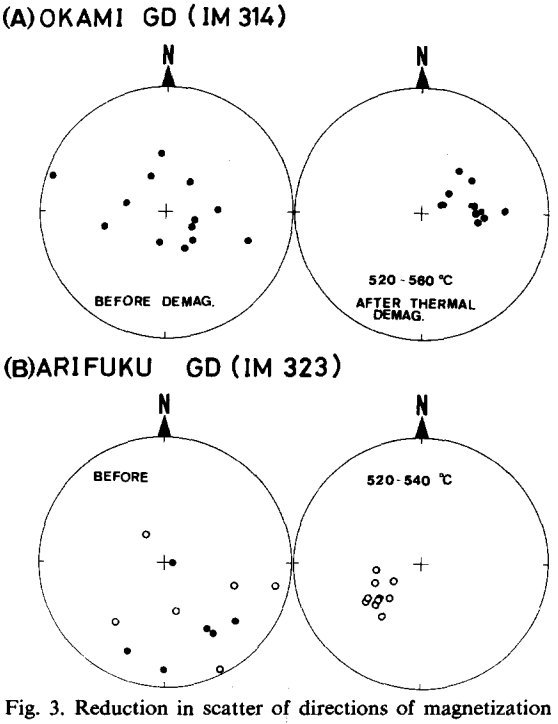


Fig. 3. Reduction in scatter of directions of magnetization by thermal demagnetization technique: A. Okami granodiorite (IM 314); B. Arifuku granodiorite (IM 323). Projections are equal area, solid symbols on the lower hemisphere, open symbols on the upper hemisphere.

(see Fig. 2), indicating that the stable portion of the remanent magnetization above 500 °C resided in titanium-poor magnetite rather than in the hematite.

Welded tuff. Except for Hamada rock unit all samples were found to be magnetically stable, and have a single component of remanent magnetization (see Fig. 2). With the aid of thermal demagnetization curves it was possible to identify a magnetic phase with a Curie temperature in the range between 580 and 680 °C in samples from four magnetically stable sites (IM 321, IM 322, IM 405, IM 406). Since little change was observed in direction of remanent magnetization after demagnetization at a Curie temperature above that of magnetite, both magnetite and hematite phases probably recorded the paleomagnetic field with almost the same direction.

The Hamada volcanic unit has recorded both normal and reversed polarities (Fig. 4). The nor-

mally magnetized sites IM 317, 413, 414 were magnetically stable and their remanent magnetization resided mainly in magnetite. In contrast, the reversely magnetized sites IM 316, IM 412 had weaker remanent magnetization in which both Curie points of magnetite and hematite were identified. Their directions of remanence migrated gradually to a shallower and more southerly direction during progressive heating up to 550 °C, and then returned to the original direction of NRM after further demagnetization above a Curie point of magnetite (see Fig. 4). This indicates that the normal component is carried by magnetite and the reversed one is by both a lower-blocking temperature phase and a hematite phase. Reflection microscopic study for the reversely magnetized samples showed that the magnetic phenocrysts were extremely altered and that the peripheral parts of magnetite phenocrysts were often altered to maghemite. As suggested by Reynolds [17] for alteration of welded tuffs, it is possible that a low-temperature phase growth (maghemitization) took place along the grain margins of magnetite. The hematite phase was probably produced during further low-temperature oxidation. Apparently, the geomagnetic field polarity changed from normal to reversed as the lower-temperature phase and the hematite phase grew and acquired remanent magnetization. Because the hematite and lower-temperature phases carried the fairly stable remanence antiparallel to the direction of magnetite phase, the data from both phases are available for paleomagnetism.

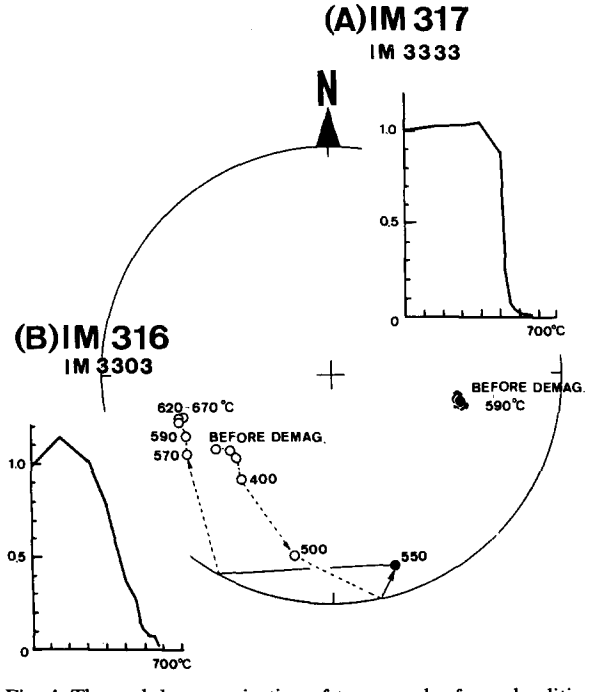


Fig. 4. Thermal demagnetization of two samples from rhyolitic welded tuff of Hamada rock unit. A. Sample from high welded portion (IM 317). Most of magnetization resides in magnetite. Only one component is observed. B. Sample from less densely welded portion (IM 316). Two components are observed. While reversed component of magnetization are present in both lower-temperature phase and hematite phase, magnetite phase has the normal component which is depressed due to presence of reversed polarity component in hematite.

3.2. Paleomagnetic data

Twenty-eight sites survived the following reliability criterion: an acceptable site has at least five specimens cut from independently oriented samples: and a mean direction from the site has a circle of 95% confidence less than 20°. The paleomagnetic data of the acceptable sites are listed in Table 1. Site mean directions were combined to obtain the mean directions of rock units listed in Table 2.

The paleomagnetic data are divided into three stages based on the fission-track ages; 30–35 Ma, 28 Ma and 21–23 Ma. Paleomagnetic directions for each stage, combined with published data [1], are drawn in Fig. 5 and listed in Table 2. The paleomagnetic field direction in Table 2 is presented as the value at a representative point (134° E, 35° N) of Southwest Japan.

30–35 Ma. The five volcanic units and two plutonic stocks have preserved a reliable paleomagnetic record. The declination (except for the Imaichi rock unit) lies inside a 46–86° range (see Fig. 1) which is concordant with the easterly declination previously reported from Southwest Japan [1].

The mean direction from the plutons can be compared with that from the other rock units; the direction from the latter is after tilting correction. Although the data from plutonic stocks have fairly steep inclinations and easterly declinations, the

TABLE 2

Average paleomagnetic directions

Stage	Rock unit	N	$\bar{D}(°)$	$\bar{I}(°)$	$\alpha_{95}(°)$	k	V.G.P.	$\partial p(°)$	$\partial m(°)$
30–35 Ma	Hamada	5	78.5	38.0	16.6	22.3			
	Hirefuriyama	4	49.1	46.6	21.2	19.8			
	Yoboshiyama	2	46.4	34.1	–	–			
	Tamagawa	1	56.9	61.6	–	–			
	Kawamoto (N) *	6	61.1	54.0	7.1	90.9			
	Kawamoto (R) *	1	64.2	44.9	–	–			
	Okami	3	64.8	60.5	16.5	56.9			
	Arifuku	4	86.3	46.3	28.1	11.7			
Mean:	sites	26	65.9	48.6	6.5	19.8	35.0 °N, 150.2 °W	5.6	8.6
Mean:	rock units	8	63.5	49.0	9.5	35.8	37.1 °N, 149.5 °W	8.2	12.6
28 Ma	Kawauchi	6	52.2	33.5	15.3	20.2	41.1 °N, 130.2 °W	9.9	17.4
21–23 Ma	Hata	6	69.9	49.5	14.5	22.2	32.2 °N, 152.7 °W	12.8	19.3

Directions are presented as the value at the representative point of Southwest Japan (134 °E, 35 °N). Data denoted by asterisk (*) are after Otofuji and Matsuda [1]. N is the number of sites, D, I are mean declination and inclination, α_{95} and k are Fisher statistic parameters, and ∂p and ∂m are semiaxes of the oval of 95% confidence around the pole. The mean directions at the stage of 30–35 Ma are calculated by assigning unit weight to rock units and sites, respectively.

circles of 95% confidence from plutonic and other rock units overlap ($D = 69.7°$, $I = 51.9°$, $\alpha_{95} = 11.8°$ vs. $D = 57.9°$, $I = 45.8°$, $\alpha_{95} = 18.8°$). This fact strongly suggests that the plutonic stocks have been subjected to little tectonic tilting and that they provide reliable paleomagnetic information. The reliability of the mean direction from eight rock units is also ascertained through the presence of normal and reversed polarity.

At sites in the Imaichi volcanic unit a partial field reversal was apparently recorded. These directions were excluded from the statistic computation for this stage.

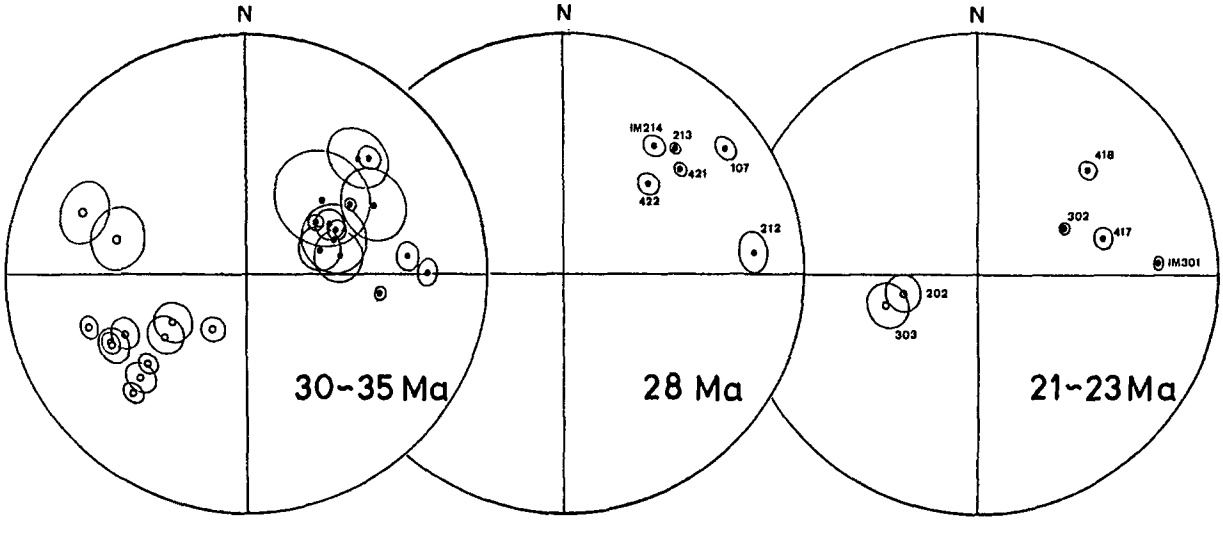


Fig. 5. Mean magnetic directions and 95% confidence limits of sites for three periods: 30–35 Ma, 28 Ma and 21–23 Ma. Projections are equal area, solid symbols on the lower hemisphere, open symbols on the upper hemisphere.

28 Ma. Two additional sites were measured for Sasahara tuff (Kawauchi cauldron) with an age of 28 Ma. These data confirm the strong easterly declination of this tuff [1].

21–23 Ma. The Hata Subgroup has layers of welded tuff to which an age of 21 Ma is assigned by fission-track dating. The magnetization of these tuffs showed normal polarity and an easterly direction with about 60° in declination. The tuff is stratigraphically sandwiched in between andesite lavas. The NRM directions from the lavas show a reversed polarity, antiparallel to that of the tuff. The mean direction of the Hata Subgroup is therefore accepted to be the average direction of the geomagnetic field over a time duration longer than the period of secular variation.

4. Discussion

4.1. Amount of rotation

Since sources of the error for paleomagnetism (tectonic tilting, overprinting of secondary component upon primary magnetization) have been carefully minimized through sampling and experimental procedures, each site datum at the stage of 30–35 Ma probably represents a spot-reading of the field direction. Data from twenty-six sites at this stage can be treated as independent. Mean direction calculated by assigning unit weight to sites is probably a realistic estimate of the paleofield for the stage of 30–35 Ma.

The remaining error source to estimate the amount of rotation of southwest Japan is the effect of the apparent polar wander (APW). To minimize this effect, the following procedure is used. The expected paleofield direction at 30 Ma is calculated for the representative point of Southwest Japan (134°E, 35°N) from the APW path of Eurasia [7]. Then its declination value is subtracted from that of the observed paleomagnetic direction. The observed paleofield direction is $D_0 = 65.9°$, $I_0 = 48.6°$ with uncertainty in declination ($\Delta D_0 = \sin^{-1}(\sin \alpha_{95}/\cos I)$) of 9.9°, while the calculated paleofield direction is $D_c = 9.7°$, $I_c = 58.9°$ and $\Delta D_c = 6.5°$. The rotation angle and its uncertainty are given by $R = D_0 - D_c = 56.2°$, $\Delta R = \sqrt{(\Delta D_0)^2 + (\Delta D_c)^2} = 11.8°$. It is concluded that Southwest Japan has undergone a clockwise rotation through 56° since 30 Ma.

A difference between the observed and calculated inclinations ($I_0 - I_c$) is $-10.3°$ with uncertainty at the 95% confidence level of $\pm 8.9°$, implying that Southwest Japan underwent northward translation. If we account for the observed inclination value calculated by assigning unit weight to rock units (see Table 2), the difference becomes $-9.9° \pm 11.2°$, which is not statistically significant. Future work can establish the significance of the northward translation of Southwest Japan with respect to the rest of Eurasia.

4.2. Timing of rotation

Approximately 60° in declination characterize the amount of rotation of Southwest Japan. A plot of declination of our study of Southwest Japan versus age (Fig. 6) clearly shows that the characteristic declination has persisted until 21 Ma, so that no clockwise rotation occurred until at least 21 Ma.

Some anomalous directions with about 60° in declination have been reported from the Miocene rocks in Southwest Japan [18–20]. The dacite welded tuffs with age of 14–18 Ma collected at Muro (136.1°E, 34.5°N) have stable magnetization with 60.1° in declination and 66.1° in inclination [21], and the dacitic rocks with age of 15 Ma at Kumano (136.0°E, 34.0°N) show 63.9° in declination and 48.4° in inclination. The anomalous directions have been attributed to either geomagnetic field reversal or the regional rotation of the blocks including these rock units. These declination values are quite consistent with the characteristic declination of Southwest Japan (see Fig. 6). Considering the frequency and duration of geomagnetic field reversals, there is little chance that two independent rock units were produced in a period of field reversal and also that they acquired a magnetization parallel to the characteristic direction of rotation of Southwest Japan during reversal. It is preferable to consider that these remanent directions reflect the tectonic rotation of Southwest Japan. These inferences suggest a

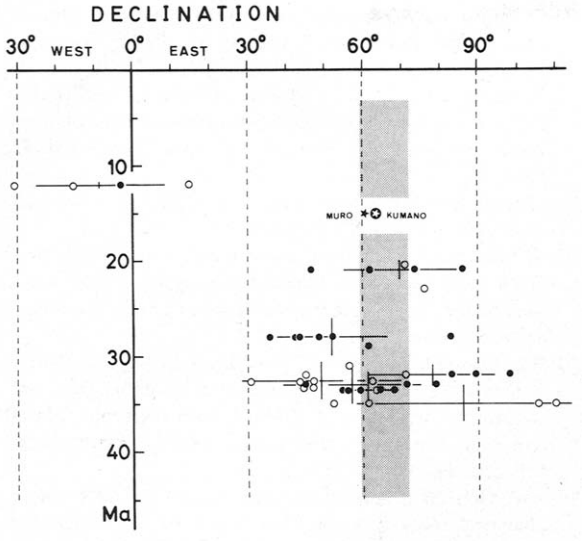


Fig. 6. Mean declination data of sites and rock units as a function of fission-track age. The declination error bars of rock unit means are the α_{95} values and age error bars are standard deviation of the average. Shaded declination zone is the declination with the α_{95} value of the mean direction of the period between 30 and 35 Ma. This value characterizes the amount of rotational motion of Southwest Japan. Closed circles show the data with normal polarity, open circles those with reversed polarity. The declination data of Muro and Kumano dacitic rocks are shown by stars.

younger origin for rotational motion. The start of rotational motion of Southwest Japan is possibly younger than 15 Ma. Taking into account that rotational motion ceased at about 12 Ma [1], the rotation through about 55° was completed within an extremely short period of 3 m.y.

4.3. Tectonic implication

Southwest Japan is bounded to the south by the Shikoku Basin whose origin is known to be back-arc spreading [22,23]. Comparing the timing of the rotation of Southwest Japan with that of the spreading of the Shikoku Basin (24 or 25 Ma to between 17 and 15 Ma [22,23]), the rotation of Southwest Japan appears to have been completed within a short span just after the Shikoku Basin had been created. The rotation model proposed by Otofuji and Matsuda [1] that Southwest Japan rotated about the pivot near its western end, combined with the timing of the rotation, indicates that the young and hot lithosphere of the Shikoku Basin was engulfed beneath the rotating Southwest Japan during fast rotation since 15 Ma. Geological evidence preserved in Southwest Japan warrants this inference. Andesitic to basaltic rocks of middle Miocene age in Southwest Japan forms the Setouchi Volcanic Belt which is arranged in a straight line nearly parallel to the Nankai Trough. The characteristics of this belt are summarized as follows [24]: (1) volcanic activity of this belt occurred within a very short time span of about 1 m.y. between 14 Ma and 13 Ma; and (2) high-magnesian andesites in this belt were generated by subduction of hot lithosphere. Both requirements for generation of this belt are satisfied by the proposed rotation model and the timing of the rotation. We therefore conclude that Southwest Japan rotated since 15 Ma, associated with the opening of the Japan Sea just after the creation of the Shikoku Basin.

The timing of rotation at about 15 Ma may be a clue to the origin and driving force for rotation of Southwest Japan. Taking into account the timing of spreading of the Parece Vela Basin (from 30 to 17 or 18 Ma [25]), the tectonic events (spreading of basins and rotation of island arc) appears to have progressed longitudinally northward from the Parece Vela Basin, through Shikoku Basin, to Southwest Japan and the Japan Sea. The longitudinal diachronism in the age of initiation of the tectonic events is also observed in the Mariana and Bonin arc systems [26,27]. This implies that the origin of the rotation of Southwest Japan is related to the phenomenon of the longitudinal diachronism of the events. The rotation of Southwest Japan is probably not only a local geological phenomenon but also the reflection of large-scale geodynamic processes which give rise to the northward migration of the tectonic events.

5. Conclusion

Based on paleomagnetic evidence it is concluded that: (1) Southwest Japan rotated clockwise 56° ± 12° during the past 30 m.y.; (2) the rotation took place suddenly and later than 15 Ma; and (3) the rotation through about 55° was com-

pleted within an extremely short period of 3 m.y. just after the Shikoku Basin had been created.

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